

Vaccine Research and Development Database System

Project Documentation

Course: CSCI204: Fundamentals of Databases

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1. Project Overview & System Scope

The Vaccine Research and Development (R&D) Database System is designed to manage complex relational data generated during clinical trials and early-stage vaccine development. Managing clinical trial workflows manually or through non-relational spreadsheets creates data fragmentation, risks participant confidentiality, and complicates adverse event tracking.

This system provides a structured MySQL relational database that coordinates candidate formulations, molecular antigens, batch manufacturing records, trial locations, trial phases, anonymized participant records, adverse reactions, and lab test results.

2. Core Business Rules

1. **Vaccine Candidates & Antigens** — Each vaccine candidate targets a specific disease and can incorporate one or more molecular antigens.
2. **Manufacturing & Batches** — Batches are strictly linked to a valid vaccine candidate. The expiration date of any given batch must be strictly later than its production date.
3. **Trial Governance & Sites** — Clinical trials take place at registered research sites across designated phases (Phase I, Phase II, or Phase III).
4. **Participant Anonymity** — Participant identities are fully masked using generated anonymized identifiers to comply with clinical trial privacy standards.
5. **Adverse Event Monitoring** — Enrolled trial participants can report multiple adverse events or none. Each record logs severity, timestamp, and clinical description.

3. Relational Schema & Entity Breakdown

Entity Definitions & Key Constraints

Table	Fields (PK / FK annotated)
vaccine_candidates	candidate_id [PK], name, technology_platform, target_disease, created_date
antigens	antigen_id [PK], candidate_id [FK], antigen_name, sequence_data
formulations	formulation_id [PK], candidate_id [FK], adjuvant_type, ph_level, storage_temp_c
batches	batch_id [PK], candidate_id [FK], batch_number, production_date, expiration_date, quantity_produced
research_sites	site_id [PK], site_name, location, contact_person

Table	Fields (PK / FK annotated)
clinical_studies	study_id [PK], candidate_id [FK], site_id [FK], phase, status, start_date, end_date
participants	participant_id [PK], anonymized_code, age, gender
study_participants	study_id [PK, FK], participant_id [PK, FK], enrollment_date, dosage_group
adverse_events	event_id [PK], participant_id [FK], study_id [FK], severity, description, report_date
lab_tests	test_id [PK], study_id [FK], participant_id [FK], test_type, result_value, test_date

Entity-Relationship Diagram (ERD)

The complete structural relationship across all 11 relational tables is illustrated below:

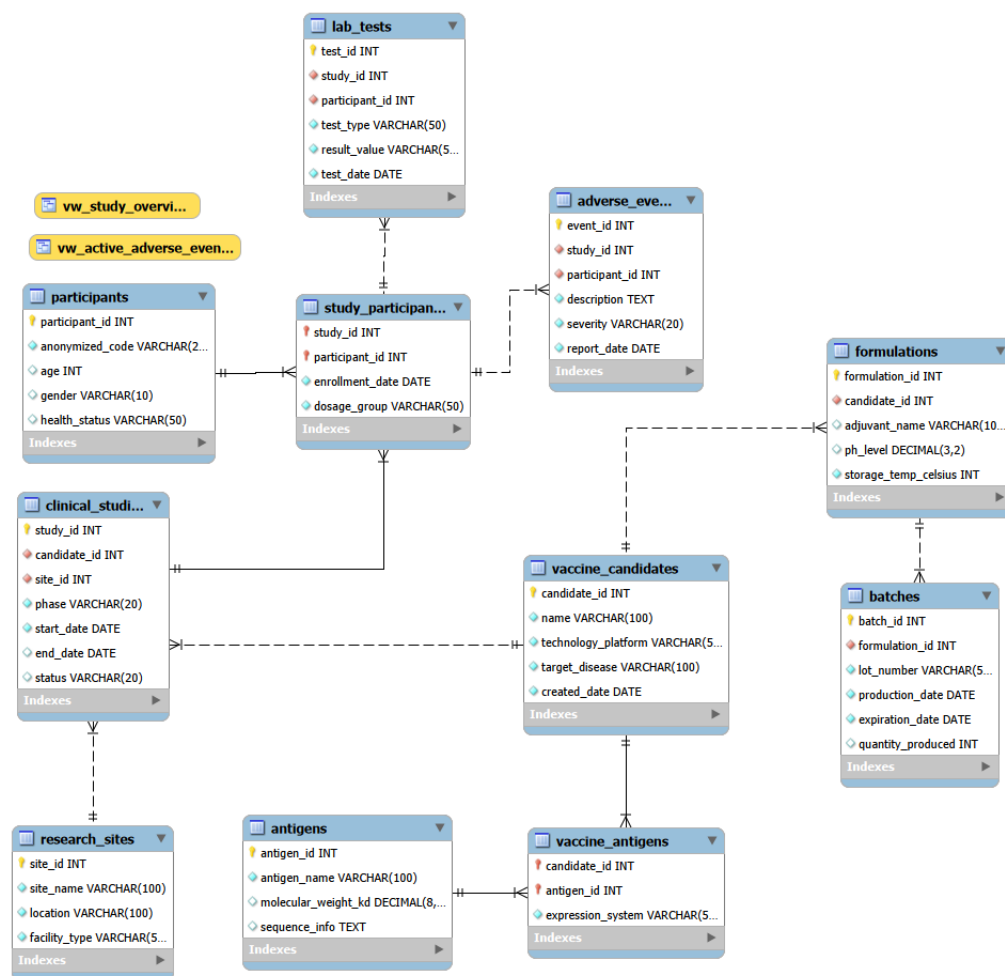


Figure 1: Complete Entity-Relationship Diagram (ERD) showing primary/foreign key mappings and cardinalities.

4. Normalization Analysis

First Normal Form (1NF)

- Every table field contains atomic values with no repeating sets or multi-valued attributes.
- Repeating attributes such as multiple lab tests or adverse event logs are stored in dedicated tables linked via Foreign Keys.

Second Normal Form (2NF)

- Meets all 1NF criteria.
- Eliminates partial key dependencies. All non-key attributes depend on the full Primary Key. For instance, in `study_participants` (composite PK: `study_id`, `participant_id`), attributes like `enrollment_date` and `dosage_group` depend strictly on both keys together.

Third Normal Form (3NF)

- Meets all 2NF criteria.
- Eliminates transitive dependencies. Attributes depend solely on the Primary Key without intermediate dependencies. Research site metadata (`location`, `contact_person`) remains inside `research_sites` rather than repeating in `clinical_studies`.

5. Database Schema Implementation

The database was instantiated by executing `create_tables.sql` followed by `load_data.sql` inside MySQL Workbench.

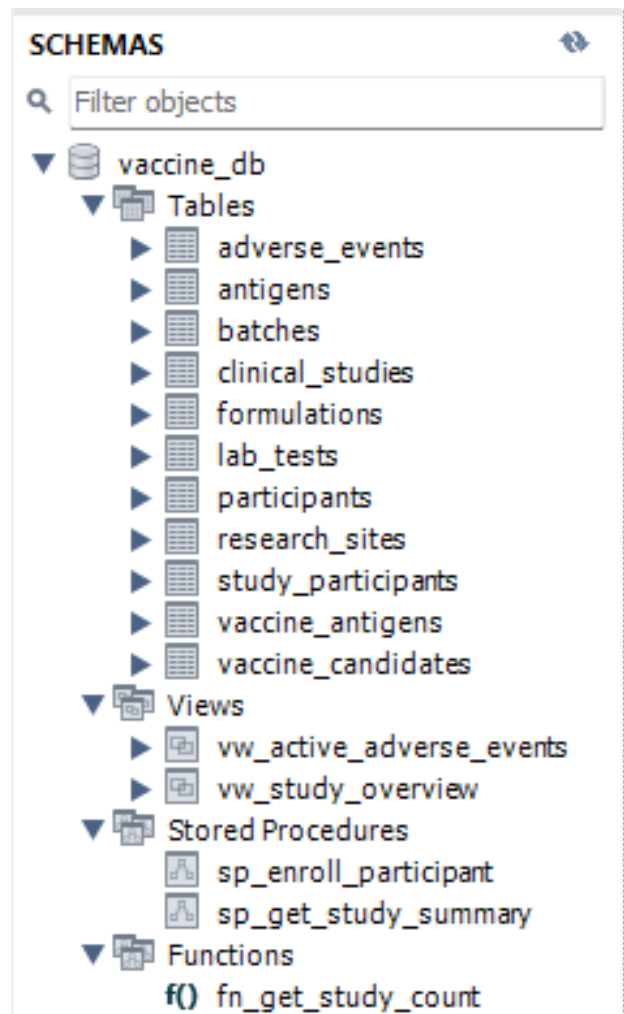


Figure 2: MySQL Workbench SCHEMAS panel displaying all 11 verified tables.

6. Analytical SQL Queries & Database Views

To evaluate research data and simplify routine reporting, custom SQL queries and relational views were defined.

6.1 Core Data Queries & Filters

1. mRNA Vaccine Candidates Filter

Queries all candidates developed using the mRNA technology platform.

```
3  -- Get mRNA vaccines
4  • SELECT candidate_id, name, target_disease, created_date
5  FROM vaccine_candidates
6  WHERE technology_platform = 'mRNA';
```

candidate_id	name	target_disease	created_date
1	Comirnaty (BNT162b2)	COVID-19	2024-01-15
2	Spikevax (mRNA-1273)	COVID-19	2024-02-10
NULL	NULL	NULL	NULL

Figure 3: Query output filtering vaccine candidates built on mRNA technology platforms.

2. Clinical Studies with Candidate & Research Site Context

Performs multi-table JOIN operations connecting trial records with candidate names and testing facility locations.

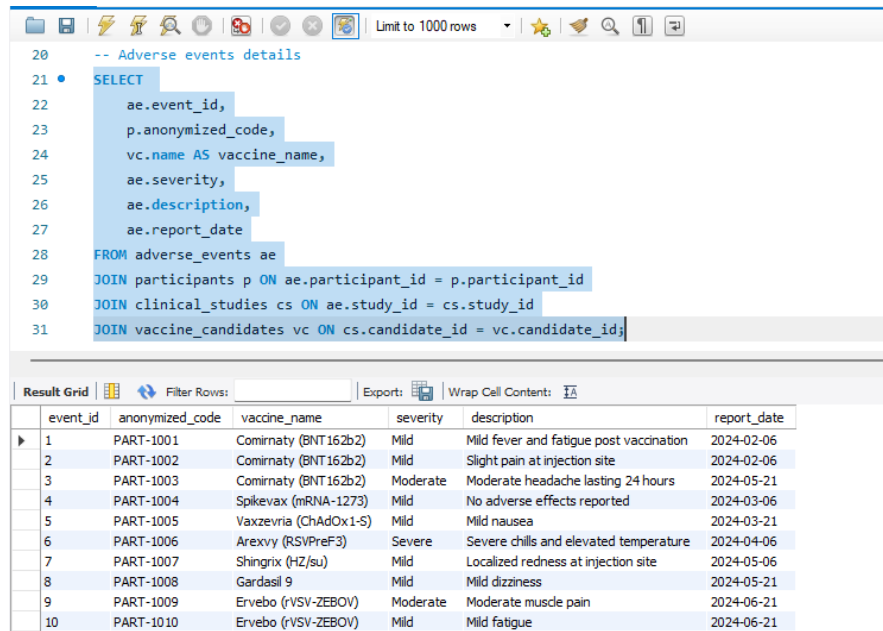
```
8  -- Studies with vaccine and site info
9  • SELECT
10     cs.study_id,
11     vc.name AS vaccine_name,
12     rs.site_name,
13     cs.phase,
14     cs.status,
15     cs.start_date
16 FROM clinical_studies cs
17 JOIN vaccine_candidates vc ON cs.candidate_id = vc.candidate_id
18 JOIN research_sites rs ON cs.site_id = rs.site_id;
```

study_id	vaccine_name	site_name	phase	status	start_date
1	Comirnaty (BNT162b2)	Kasr Al Ainy Hospital - Cairo University	Phase I	Completed	2024-02-01
2	Comirnaty (BNT162b2)	Ain Shams University Hospitals	Phase II	Completed	2024-05-15
3	Spikevax (mRNA-1273)	Theodor Bilharz Research Institute (TBRI)	Phase I	Completed	2024-03-01
4	Vaxzevria (ChAdOx1-S)	National Research Centre (NRC)	Phase I	Ongoing	2024-03-15
5	Arexvry (RSVPreF3)	Mansoura University Main Hospital	Phase I	Ongoing	2024-04-01
6	Mosquirix (RTS,S/AS01)	Alexandria University Main Hospital	Phase I	Pending	2024-04-20
7	Shingrix (HZ/su)	Suez Canal University Hospital	Phase II	Ongoing	2024-05-01
8	Gardasil 9	Assiut University Main Hospital	Phase I	Ongoing	2024-05-15
9	Engerix-B	Tanta University Hospital	Phase I	Pending	2024-06-01
10	Ervebo (rVSV-ZEBOV)	Childrens Cancer Hospital 57357	Phase III	Ongoing	2024-06-15

Figure 4: Relational JOIN output linking clinical studies, site names, and vaccine candidates.

3. Detailed Adverse Event Logs

Fetches adverse reaction records combined with trial phase metadata and anonymized participant profiles.



The screenshot shows a SQL query editor with a query to fetch adverse event details. The query selects event_id, anonymized_code, vaccine_name, severity, description, and report_date from the adverse_events table, joined with participants, clinical_studies, and vaccine_candidates tables. The results grid displays 10 rows of data.

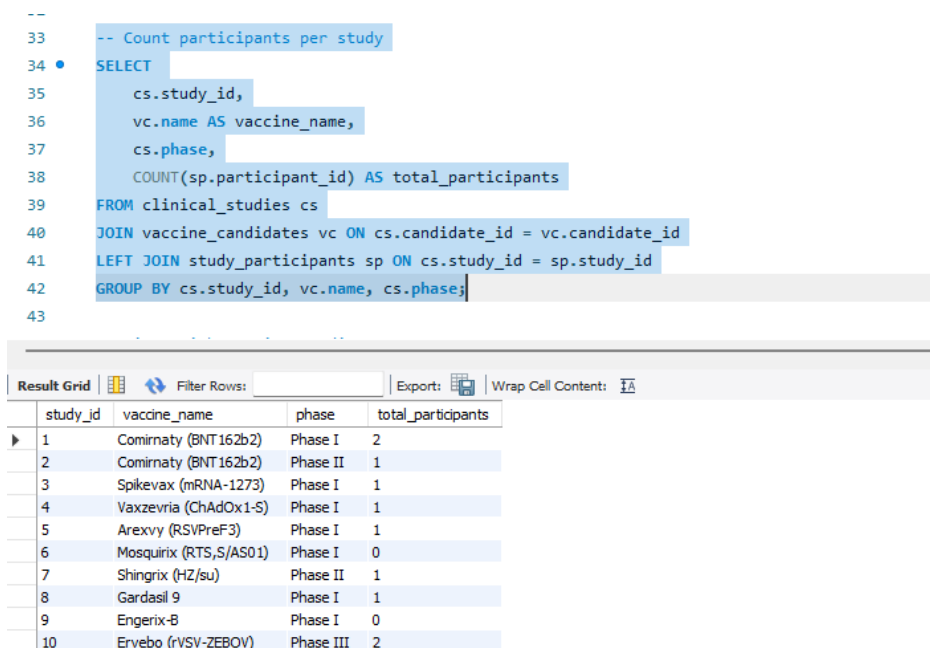
```
-- Adverse events details
SELECT
    ae.event_id,
    p.anonymized_code,
    vc.name AS vaccine_name,
    ae.severity,
    ae.description,
    ae.report_date
FROM adverse_events ae
JOIN participants p ON ae.participant_id = p.participant_id
JOIN clinical_studies cs ON ae.study_id = cs.study_id
JOIN vaccine_candidates vc ON cs.candidate_id = vc.candidate_id;
```

event_id	anonymized_code	vaccine_name	severity	description	report_date
1	PART-1001	Comirnaty (BNT162b2)	Mild	Mild fever and fatigue post vaccination	2024-02-06
2	PART-1002	Comirnaty (BNT162b2)	Mild	Slight pain at injection site	2024-02-06
3	PART-1003	Comirnaty (BNT162b2)	Moderate	Moderate headache lasting 24 hours	2024-05-21
4	PART-1004	Spikevax (mRNA-1273)	Mild	No adverse effects reported	2024-03-06
5	PART-1005	Vaxzevria (ChAdOx1-S)	Mild	Mild nausea	2024-03-21
6	PART-1006	Arexvry (RSVPreF3)	Severe	Severe chills and elevated temperature	2024-04-06
7	PART-1007	Shingrix (HZ/su)	Mild	Localized redness at injection site	2024-05-06
8	PART-1008	Gardasil 9	Mild	Mild dizziness	2024-05-21
9	PART-1009	Ervebo (rVSV-ZEBOV)	Moderate	Moderate muscle pain	2024-06-21
10	PART-1010	Ervebo (rVSV-ZEBOV)	Mild	Mild fatigue	2024-06-21

Figure 5: Detailed adverse events query displaying reaction severity and participant codes.

4. Participant Enrollment Aggregation per Trial

Utilizes GROUP BY and COUNT() functions to summarize total participant counts across active studies.



The screenshot shows a SQL query to count participants per study. The query selects study_id, vaccine_name, phase, and total_participants (COUNT(sp.participant_id)) from the clinical_studies table, joined with vaccine_candidates and study_participants tables. The results grid displays 10 rows of data.

```
-- Count participants per study
SELECT
    cs.study_id,
    vc.name AS vaccine_name,
    cs.phase,
    COUNT(sp.participant_id) AS total_participants
FROM clinical_studies cs
JOIN vaccine_candidates vc ON cs.candidate_id = vc.candidate_id
LEFT JOIN study_participants sp ON cs.study_id = sp.study_id
GROUP BY cs.study_id, vc.name, cs.phase;
```

study_id	vaccine_name	phase	total_participants
1	Comirnaty (BNT162b2)	Phase I	2
2	Comirnaty (BNT162b2)	Phase II	1
3	Spikevax (mRNA-1273)	Phase I	1
4	Vaxzevria (ChAdOx1-S)	Phase I	1
5	Arexvry (RSVPreF3)	Phase I	1
6	Mosquirix (RTS,S/AS01)	Phase I	0
7	Shingrix (HZ/su)	Phase II	1
8	Gardasil 9	Phase I	1
9	Engerix-B	Phase I	0
10	Ervebo (rVSV-ZEBOV)	Phase III	2

Figure 6: Participant aggregation per clinical trial phase.

5. Research Sites Hosting Active Studies

Identifies physical testing locations currently managing active clinical trials.

```
44  -- Sites with ongoing studies
45  • SELECT
46      rs.site_name,
47      rs.location,
48      COUNT(cs.study_id) AS total_studies
49  FROM research_sites rs
50  JOIN clinical_studies cs ON rs.site_id = cs.site_id
51  GROUP BY rs.site_id, rs.site_name, rs.location
52  HAVING COUNT(cs.study_id) >= 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
site_name	location	total_studies	
Kasr Al Ainy Hospital - Cairo University	Cairo, Egypt	1	
Ain Shams University Hospitals	Cairo, Egypt	1	
Theodor Bilharz Research Institute (TBRI)	Giza, Egypt	1	
National Research Centre (NRC)	Giza, Egypt	1	
Mansoura University Main Hospital	Mansoura, Egypt	1	
Alexandria University Main Hospital	Alexandria, Egypt	1	
Suez Canal University Hospital	Ismailia, Egypt	1	
Assiut University Main Hospital	Assiut, Egypt	1	
Tanta University Hospital	Tanta, Egypt	1	
Childrens Cancer Hospital 57357	Cairo, Egypt	1	

Figure 7: Active research site locations and assigned clinical trials.

6. High-Severity Side Effect Reporting

Filters participant logs specifically for reactions flagged with severe classification levels.

```
53
54  -- Participants with severe side effects
55  • SELECT participant_id, anonymized_code, age, gender
56  FROM participants
57  WHERE participant_id IN (
58      SELECT participant_id
59      FROM adverse_events
60      WHERE severity = 'Severe'
61  );
```

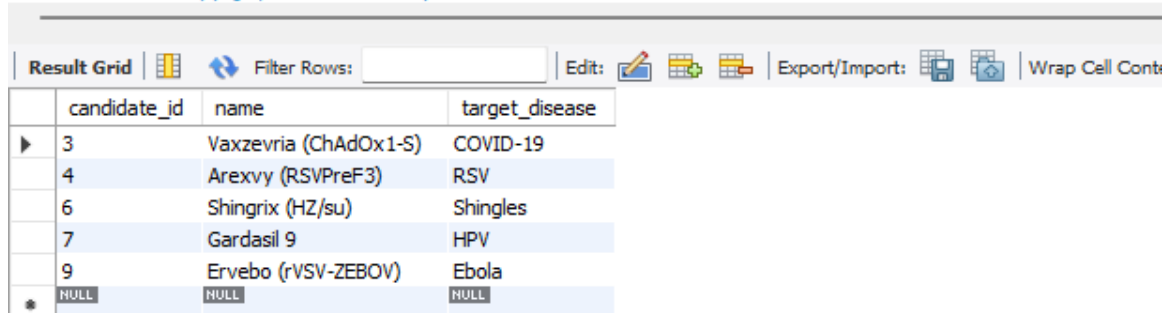
Result Grid	Filter Rows:	Edit:	Export/Import:
participant_id	anonymized_code	age	gender
6	PART-1006	38	Female
* NULL	NULL	NULL	NULL

Figure 8: Filtered report showing severe adverse event records.

7. Active Vaccine Trials Summary

Displays candidates currently undergoing active Phase I, II, or III evaluation.

```
63  -- Vaccines with ongoing trials
64  • SELECT candidate_id, name, target_disease
65  FROM vaccine_candidates vc
66  WHERE EXISTS (
67      SELECT 1
68      FROM clinical_studies cs
69      WHERE cs.candidate_id = vc.candidate_id
70            AND cs.status = 'Ongoing'
71  );
72
```



The screenshot shows a database query result grid with the following data:

candidate_id	name	target_disease
3	Vaxzevria (ChAdOx1-S)	COVID-19
4	Arexvy (RSVPreF3)	RSV
6	Shingrix (HZ/su)	Shingles
7	Gardasil 9	HPV
9	Ervebo (rVSV-ZEBOV)	Ebola
NULL	NULL	NULL

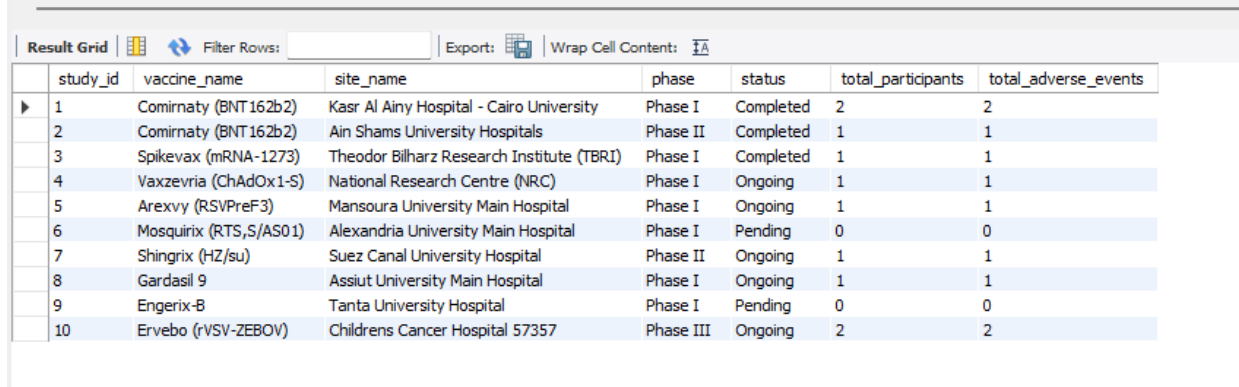
Figure 9: Summary query of candidate vaccines currently in active testing.

6.2 Pre-compiled Relational Views

1. Study Overview View (vw_study_overview)

Combines trial metrics, candidate details, site details, total enrolled participants, and reported event counts into a single view.

```
37  -- Testing the views
38  • SELECT * FROM vw_study_overview;
```



The screenshot shows a database query result grid with the following data:

study_id	vaccine_name	site_name	phase	status	total_participants	total_adverse_events
1	Comirnaty (BNT162b2)	Kasr Al Ainy Hospital - Cairo University	Phase I	Completed	2	2
2	Comirnaty (BNT162b2)	Ain Shams University Hospitals	Phase II	Completed	1	1
3	Spikevax (mRNA-1273)	Theodor Bilharz Research Institute (TBRI)	Phase I	Completed	1	1
4	Vaxzevria (ChAdOx1-S)	National Research Centre (NRC)	Phase I	Ongoing	1	1
5	Arexvy (RSVPreF3)	Mansoura University Main Hospital	Phase I	Ongoing	1	1
6	Mosquirix (RTS,S/AS01)	Alexandria University Main Hospital	Phase I	Pending	0	0
7	Shingrix (HZ/su)	Suez Canal University Hospital	Phase II	Ongoing	1	1
8	Gardasil 9	Assiut University Main Hospital	Phase I	Ongoing	1	1
9	Engerix-B	Tanta University Hospital	Phase I	Pending	0	0
10	Ervebo (rVSV-ZEBOV)	Childrens Cancer Hospital 57357	Phase III	Ongoing	2	2

Figure 10: Execution output of SELECT * FROM vw_study_overview.

2. Active Adverse Events View (vw_active_adverse_events)

Provides a clinical safety stream linking anonymized patient IDs, age, gender, severity, and vaccine names.

```
37 -- Testing the views
38 • SELECT * FROM vw_study_overview;
39 • SELECT * FROM vw_active_adverse_events;
```

	event_id	anonymized_code	age	gender	vaccine_name	phase	severity	description	report_date
▶	1	PART-1001	24	Male	Comirnaty (BNT162b2)	Phase I	Mild	Mild fever and fatigue post vaccination	2024-02-06
	2	PART-1002	31	Female	Comirnaty (BNT162b2)	Phase I	Mild	Slight pain at injection site	2024-02-06
	3	PART-1003	45	Male	Comirnaty (BNT162b2)	Phase II	Moderate	Moderate headache lasting 24 hours	2024-05-21
	4	PART-1004	29	Female	Spikevax (mRNA-1273)	Phase I	Mild	No adverse effects reported	2024-03-06
	5	PART-1005	52	Male	Vaxzevria (ChAdOx1-S)	Phase I	Mild	Mild nausea	2024-03-21
	6	PART-1006	38	Female	Arexvy (RSVPreF3)	Phase I	Severe	Severe chills and elevated temperature	2024-04-06
	7	PART-1007	22	Male	Shingrix (HZ/su)	Phase II	Mild	Localized redness at injection site	2024-05-06
	8	PART-1008	60	Female	Gardasil 9	Phase I	Mild	Mild dizziness	2024-05-21
	9	PART-1009	27	Male	Ervebo (rVSV-ZEBOV)	Phase III	Moderate	Moderate muscle pain	2024-06-21
	10	PART-1010	35	Female	Ervebo (rVSV-ZEBOV)	Phase III	Mild	Mild fatigue	2024-06-21

Figure 11: Execution output of `SELECT * FROM vw_active_adverse_events`.

7. Stored Procedures & Logic Implementation

Business logic is encapsulated in database procedures for consistency. The procedure `sp_get_study_summary` takes a target `study_id` as input and computes trial statistics dynamically.

```
37
38 • -- Test Execution
39 CALL sp_get_study_summary(1);
```

	study_id	vaccine_name	phase	status	total_participants
▶	1	Comirnaty (BNT162b2)	Phase I	Completed	2

Figure 12: Execution of stored procedure `CALL sp_get_study_summary(1)` in MySQL Workbench.

8. Web Application & Database Integration (Streamlit UI)

A web interface built with Python and Streamlit (src/app.py) provides graphical access to the underlying MySQL database for CRUD operations.

8.1 Interface Navigation & Table Browsing

Users can select tables or views from the interface sidebar and execute real-time search queries across records.

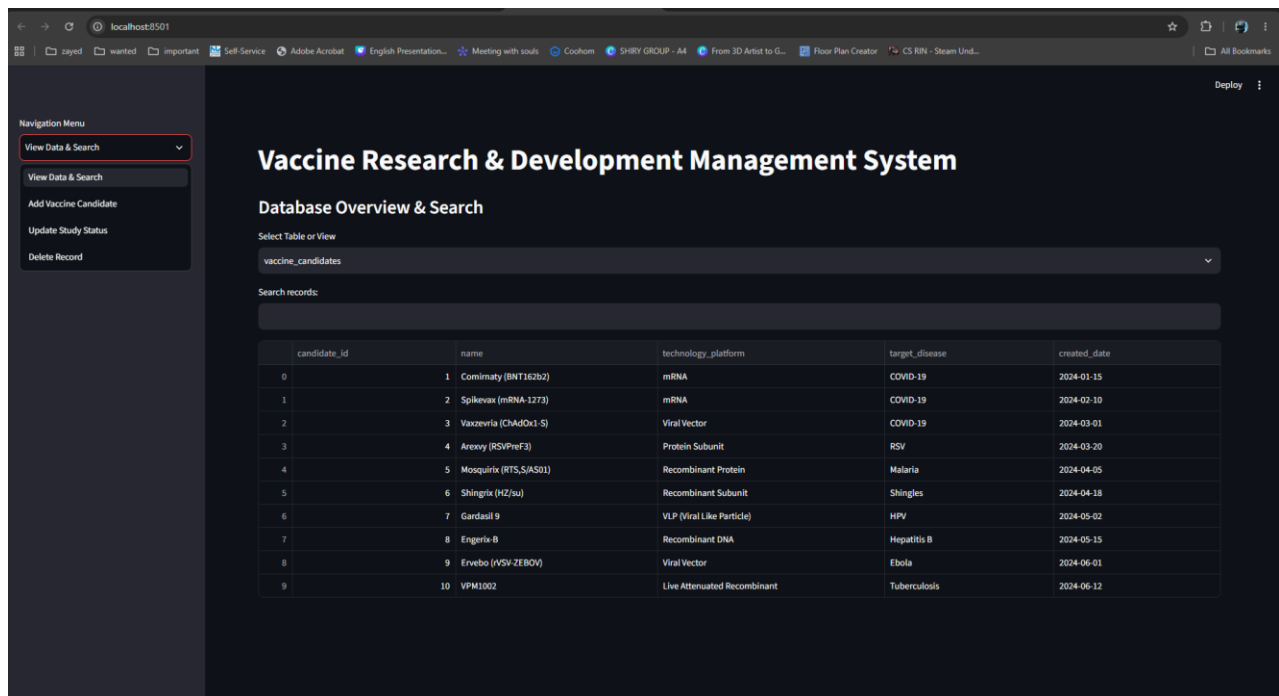
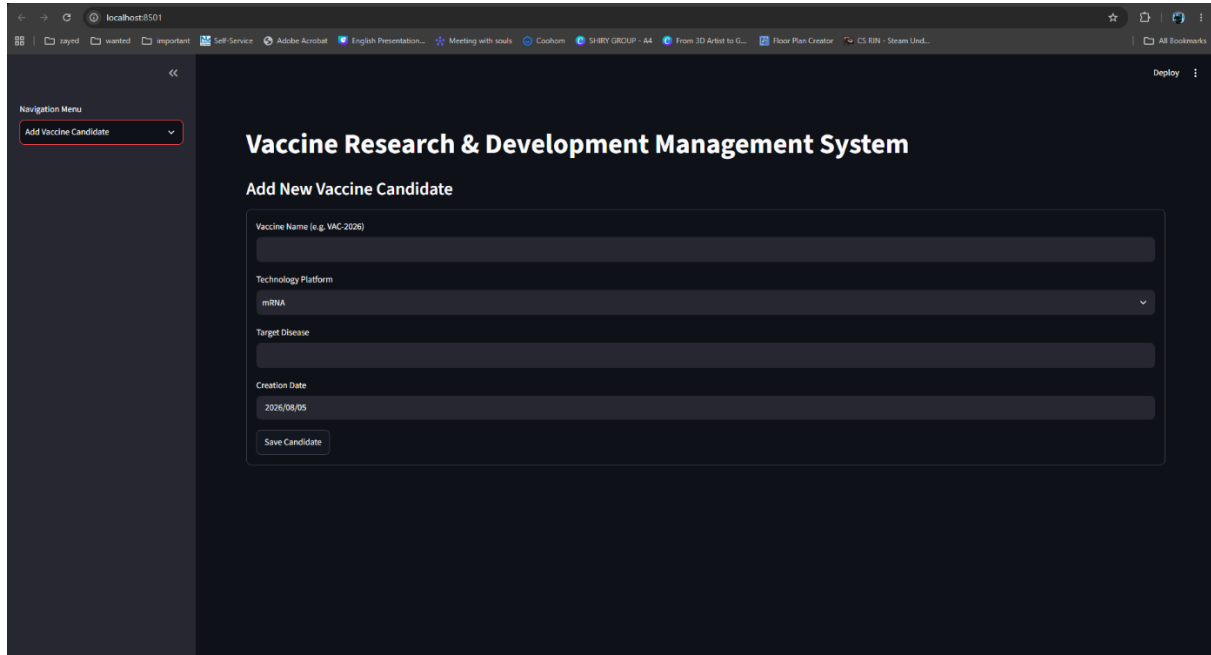


Figure 13: Streamlit interface database overview with navigation options for searching, adding, updating, and deleting records.

8.2 End-to-End Data Insertion Workflow

Step 1. Inputting New Vaccine Candidate Data

The user accesses the "Add Vaccine Candidate" form to register a new candidate named vaccine_example.



The screenshot shows a web browser window with the URL localhost:8501. The page title is "Vaccine Research & Development Management System". On the left, a navigation menu has "Add Vaccine Candidate" selected. The main content area is titled "Add New Vaccine Candidate" and contains a form with the following fields: "Vaccine Name (e.g. VAC-2026)" (text input), "Technology Platform" (dropdown menu showing "mRNA"), "Target Disease" (text input), and "Creation Date" (text input showing "2026/08/05"). A "Save Candidate" button is at the bottom of the form.

Figure 14: Interactive data input form for adding a new vaccine candidate.

Step 2. Form Submission & Confirmation

Upon submitting valid data, the application executes an SQL INSERT query and confirms completion.



This screenshot shows the same form as Figure 14, but with data entered: "vaccine_example" in the Vaccine Name field, "Viral Vector" in the Technology Platform dropdown, "TEST" in the Target Disease field, and "2026/08/05" in the Creation Date field. The "Save Candidate" button is highlighted with a red border. Below the form, a green message box with a red border displays the text: "Vaccine Candidate 'vaccine_example' added successfully."

Figure 15: System confirmation prompt verifying successful record insertion.

Step 3. Verifying Database Persistence

Navigating back to the main overview confirms that vaccine_example is stored in the MySQL database.

Vaccine Research & Development Management System					
Database Overview & Search					
Select Table or View					
vaccine_candidates					
Search records:					
	candidate_id	name	technology_platform	target_disease	created_date
1	2	Spikevax (mRNA-1273)	mRNA	COVID-19	2024-02-10
2	3	Vaxzevria (ChAdOx1-S)	Viral Vector	COVID-19	2024-03-01
3	4	Arexvy (RSVPref3)	Protein Subunit	RSV	2024-03-20
4	5	Mosquirix (RTS,S/AS01)	Recombinant Protein	Malaria	2024-04-05
5	6	Shingrix (HZ/su)	Recombinant Subunit	Shingles	2024-04-18
6	7	Gardasil 9	VLP (Viral Like Particle)	HPV	2024-05-02
7	8	Engerix-B	Recombinant DNA	Hepatitis B	2024-05-15
8	9	Ervebo (rVSV-ZEBOV)	Viral Vector	Ebola	2024-06-01
9	10	VPM1002	Live Attenuated Recombinant	Tuberculosis	2024-06-12
10	11	vaccine_example	Viral Vector	TEST	2026-08-05

Figure 16: Updated database table view reflecting the newly inserted candidate record.